

Computer Networks Cheat Sheet

1. Internet Design Goals

Primary Goal

Integrate several separately administrated entities into a common entity

Secondary Goals

- **Survivability:** Must continue despite loss of network or gateway
- **Multiple services:** Support multiple types of services/applications
- **Multiple networks:** Support multiple types of networks
- **Distributed management:** Decentralized resource management
- **Cost effective:** Scalable architecture
- **Easy attachment:** Low barrier for hosts to join
- **Accountability:** Resources must be accountable

2. Circuit vs Packet Switching

Circuit Switching

- Dedicated route between source and destination
- Single stream of bytes per route
- Uses TDM/FDM/CDM multiplexing
- Resource reservation for entire session
- Example: Traditional telephone networks

Packet Switching

- Data divided into packets sent individually
- Medium occupied only during packet transmission
- Statistical multiplexing - shares bandwidth dynamically
- Each packet is self-contained with source and destination address
- Independently routed from source to destination
- “Best effort” approach - no guarantees
- Each machine sends packet to router believed closest to destination

Comparison

Circuit: Dedicated bandwidth, setup required, guaranteed performance, fails on failure. Packet: Shared bandwidth, no setup, best-effort, reroutable, higher efficiency for bursty data.

3. Protocols

Protocol Definition

A protocol defines:

- Roles of communicating entities
- Format of messages
- Order of messages
- Actions taken on transmission, receipt of message, or other event

A fully-defined protocol must provide proper action for any event in any state

Internet Protocols

- **Client:** Party that initiates conversation
- **Server:** Party that passively waits for conversations
- Many are request-response protocols
- Well-defined rules for whose turn it is

4. Internet Protocol Stack

Application: HTTP, SMTP, FTP, DNS. Unit: Message. Network applications.

Transport: TCP (reliable), UDP (unreliable). Unit: Segment. Identifies process, ensures order (TCP), recovers loss (TCP).

Network: IP, ICMP, OSPF, BGP. Unit: Packet. Routes through routers to destination.

Link: Ethernet, WiFi. Unit: Frame. Routes to adjacent machines.

Physical: 802.11, 1000BASE-T. Unit: Bits. Encodes for medium.

Encapsulation (sending): App creates message → Transport adds header (ports) → Network adds IP header (IPs) → Link adds Ethernet header/trailer (MACs, CRC) → Physical converts to bits.

At Router: Removes link header, examines IP header only, routes on dest IP, adds new link header. Does NOT examine transport/app layers.

At Destination: De-encapsulate layer by layer, each processes its header, deliver to app.

5. IP Addresses

IP Address Basics

- Every machine reachable on Internet has at least one IP address
- Technically, each interface has at least one
- Internally: sequence of 32 bits (IPv4)
- Human format: 4 8-bit unsigned integers a.b.c.d
- Each of a, b, c, d is 0-255

Example:

```
11010110 10110110 00011001 00010111
214 . 182 . 25 . 23
```

IP Addresses and Routing

- Sender sends message to router/gateway
- Router decides where to send packet based on destination IP
- Routers need to know how to get to each possible destination
- Question: How many mappings does router need? How much memory?

6. Autonomous Systems (AS)

Internet organized into ASs (remnant of private networks). Each AS: assigned IP range, routes to owned addresses and non-owned. Some do end systems (last mile), some do backbones (long haul), some both, some consortiums. Networks connect at Internet Exchanges.

Peering: Two ASs pass traffic between customers. **Transit:** AS passes traffic from source AS to different AS.

7. CIDR - Classless Inter-Domain Routing

Purpose

- AS receives range of addresses
- All routing from outside AS only needs to reach the AS
- Routing table does not need to list every single IP address
- Routing information propagation (announcement) compacted
- AS takes care of internal routing

CIDR Prefix Notation

All addresses in range have same fixed bits

Format: First address in range / number of fixed bits

- Fixed bits = network address
- Unfixed bits = host address

Example: 123.123.96.0/19

```
123.123.96.0 =
01111011 01111011 01100000 00000000
Network (19 bits): 01111011 01111011 011
Host (13 bits): 00000 00000000
```

Total addresses in range: $2^{13} = 8192$
First address (network): 123.123.96.0
Last address (broadcast): 123.123.127.255

Netmask Alternative Format

CIDR /19 = Netmask with first 19 bits as 1:

```
11111111 11111111 11100000 00000000
255 . 255 . 224 . 0
```

Checking Same Network

To check if two IPs in same network:

1. Apply netmask to both IPs using bitwise AND
2. If results equal, IPs are on same network

Example: Host IP 192.168.45.24/255.255.240.0

Destination IP 192.168.49.104

Are they on same network?

```
192.168.45.24 = 11000000 10101000 00101101 00011000
255.255.240.0 = 11111111 11111111 11110000 00000000
Network addr = 11000000 10101000 00100000 00000000
```

```
192.168.49.104 = 11000000 10101000 00110001 01101000
255.255.240.0 = 11111111 11111111 11110000 00000000
Network addr = 11000000 10101000 00110000 00000000
```

Different network addresses -> NOT same network

Key Formulas

Given CIDR /n:

$$\begin{aligned} \text{Host bits} &= 32 - n \\ \text{Total addresses} &= 2^{32-n} \\ \text{Usable hosts} &= 2^{32-n} - 2 \end{aligned}$$

8. Routing

Forwarding Table (FIB): Router keeps table linking IP range to link/next hop. Built from routing table (RIB).

Longest Prefix Match: Multiple ranges match → use most specific (longest prefix). Example: 192.168.35.0/24 more specific than 192.168.0.0/16.

AS Routing: Each AS tells others which addresses it handles and routes to.

EBGP: Advertises prefixes with AS number, AS path (prevents loops), next hop. Receiver builds table using: AS decisions, shortest route, hot potato routing.

Within AS - iBGP: External routers communicate about external routes. **IGPs:** OSPF (link-state, topology-based), RIP (distance-vector, hop count-based).

9. IP Address Assignment

Company joining ISP receives IP range from ISP. Alternative: keep IP from previous ISP (ISP announces prefix). ISP gets from RIR (Regional Internet Registry - US/Canada: ARIN). RIRs get from IANA. Network admins divide into subnets for departments/labs. Servers/desktops may have fixed manual IPs.

Problems: Address changes for subnet rearrangement. Mobile devices need different IPs in different networks. **Solution:** DHCP - computers ask for IP when connecting.

10. DHCP - Dynamic Host Configuration Protocol

DHCP Concept

- Host needing IP address requests one from server
- What server? Send broadcast request

DHCP Steps (DORA)

1. DHCP Discover:

- Client asks server for IP address
- Sent as broadcast

2. DHCP Offer:

- Server provides possible IP address to client

3. DHCP Request:

- Client accepts offer and requests to be assigned IP address

4. DHCP ACK:

- Server acknowledges request and assigns new IP address to client

DHCP Offer/ACK Includes

- Unique IP address for client
- Netmask for local network
- Lease time: How long can you use IP before renewing (in seconds)
- Routing information: Usually IP address for only one catch-all router (default gateway)
- Host name, domain name (optional)
- Name (DNS) server (optional)

11. Traceroute

Tool to determine path to destination. Sends packets hoping for errors (interested in errors, not packet). Returning errors tell which router returned it and how long. Doesn't always work. May find multiple routes. Reverse route usually assumed same unless told otherwise. Router IPs may differ on two paths (routers have multiple interfaces/IPs).

Congestion signs: Sudden RTT increase at hop, packet loss (***), variable RTTs, persisting increase. **High latency:** Large RTT jumps between hops (geographic links). Link latency = RTT(n) - RTT(n-1).

12. Practice Problems

Problem 1: Network Layer Abstraction

Which statements are true about network layer abstraction?

(a) As message traverses router towards destination, router will use (and modify) information in network layer (IP) header

TRUE - Routers operate at network layer. They examine and modify IP headers (e.g., decrement TTL, recalculate checksum, may fragment packets).

(b) Program using socket to send messages needs to implement application layer protocol's rules in program itself

TRUE - Application developer must implement application protocols (HTTP, FTP, etc.) in their program. The OS provides socket interface to transport layer, but application logic is developer's responsibility.

(c) As message reaches physical layer, it contains at least three identifications: port number (transport), IP address (network), and MAC address (link)

TRUE - Through encapsulation, when packet reaches physical layer, it has been wrapped with headers from each layer: transport header (with ports), network header (with IP addresses), and link header (with MAC addresses). All are present in transmitted frame.

(d) Program using socket needs to implement transport layer protocol tasks

FALSE - Transport layer (TCP/UDP) is handled by operating system, not application program. Applications use sockets which provide interface to transport layer. OS handles reliability, flow control, etc.

Problem 2: CIDR for 24 Computers

Network admin needs CIDR range for 24 computers. Which IP ranges are sufficient?

Analysis:

- Need 24 hosts + 2 (network + broadcast) = 26 total addresses
- Minimum: 2⁵ = 32 addresses
- Requires CIDR /27 or larger block (smaller number after /)

Options:

- (a) 12.16.21.192/26: 2³²⁻²⁶ = 2⁶ = 64 addresses ✓
- (b) 18.85.129.0/27: 2³²⁻²⁷ = 2⁵ = 32 addresses ✓
- (c) 143.3.90.12/30: 2³²⁻³⁰ = 2² = 4 addresses ×
- (d) 73.29.127.0/25: 2³²⁻²⁵ = 2⁷ = 128 addresses ✓
- (e) 117.24.227.56/29: 2³²⁻²⁹ = 2³ = 8 addresses ×

(f) 210.142.188.176/28: 2³²⁻²⁸ = 2⁴ = 16 addresses ×

Answers: (a), (b), (d)

Problem 3: CIDR Block 33.35.125.5/27

Given CIDR block: 33.35.125.5/27

Part A: What is the range of addresses?

Understanding: /27 means first 27 bits are network, last 5 are host.

Host bits = 32 - 27 = 5 bits → Total addresses = 2⁵ = 32

Convert IP to binary:

33 = 00100001, 35 = 00100011
125 = 01111101, 5 = 00000101
Full: 00100001.00100011.01111101.00000101

Apply /27 mask - first 27 bits are network:

00100001.00100011.01111101.000|00101
Network: 00100001.00100011.01111101.000
Host: 00101

Network address (all host bits = 0):

00100001.00100011.01111101.00000000 = 33.35.125.0

Broadcast address (all host bits = 1):

00100001.00100011.01111101.00011111 = 33.35.125.31

Verify: 5 host bits give values 0-31 in last octet (00000 to 11111), so addresses run from 33.35.125.0 to 33.35.125.31.

Answer Part A: 33.35.125.0 to 33.35.125.31 (32 addresses total)

Part B: Network mask in dotted-decimal?

Method: For /27, first 27 bits = 1, last 5 bits = 0

Build binary mask:

Octet 1: 11111111 (8 ones) = 255
Octet 2: 11111111 (8 ones) = 255
Octet 3: 11111111 (8 ones) = 255
Octet 4: 11100000 (3 ones, 5 zeros)

Convert octet 4: 11100000 = 128+64+32 = 224

Answer Part B: 255.255.255.224

Verify by applying mask to IP:

IP: 00100001.00100011.01111101.00000101
Mask: 11111111.11111111.11111111.11100000
AND = 00100001.00100011.01111101.00000000
= 33.35.125.0 (network address)

Part C: How many hosts can be configured?

Calculate total: 5 host bits → 2⁵ = 32 total addresses

Subtract reserved: Two addresses cannot be assigned to hosts:

- 33.35.125.0 (network address - identifies the network)
- 33.35.125.31 (broadcast - sends to all hosts)

Usable hosts: 32 - 2 = 30

Valid range: First: 33.35.125.1, Last: 33.35.125.30

Answer Part C: 30 hosts

General formula for CIDR /n: Usable = 2⁽³²⁻ⁿ⁾ - 2
For /27: 2⁽³²⁻²⁷⁾ - 2 = 2⁵ - 2 = 30